



Java HashSet

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Java HashSet

A **HashSet** is a collection of elements where every element is **unique**.

It is part of the **java.util** package and implements the **Set** interface.

Create a HashSet

Example

Create a **HashSet** object called **cars** that will store strings:

```
import java.util.HashSet; // Import the HashSet class

HashSet<String> cars = new HashSet<String>();
```

Now you can use methods like **add()**, **contains()**, and **remove()** to manage your collection of unique elements.



Example

```
// Import the HashSet class
import java.util.HashSet;

public class Main {
    public static void main(String[] args) {
        HashSet<String> cars = new HashSet<String>();
        cars.add("Volvo");
        cars.add("BMW");
        cars.add("Ford");
        cars.add("BMW"); // Duplicate
        cars.add("Mazda");
        System.out.println(cars);
    }
}
```

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Note: In the example above, even though "BMW" is added twice, it only appears once in the set because sets do not allow duplicate elements.

Check If an Element Exists

To check whether an element exists in a `HashSet`, use the `contains()` method:

Example



Remove an Element

To remove an element, use the `remove()` method:

Example

```
cars.remove("Volvo");
```

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To remove all elements, use the `clear()` method:

Example

```
cars.clear();
```

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HashSet Size



```
cars.size();
```

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Note: Duplicate values are not counted - only unique elements are included in the size.

Loop Through a HashSet

Loop through the elements of an `HashSet` with a **for-each** loop:

Example

```
for (String i : cars) {  
    System.out.println(i);  
}
```

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Other Types

Elements in an `HashSet` are actually objects. In the examples above, we created elements (objects) of type "String". Remember that a `String` in Java is an object (not a primitive type). To use other types, such as `int`, you must specify an equivalent wrapper class: `Integer`. For other primitive types, use: `Boolean` for boolean, `Character` for char, `Double` for double, etc:



```
import java.util.HashSet;

public class Main {
    public static void main(String[] args) {

        // Create a HashSet object called numbers
        HashSet<Integer> numbers = new HashSet<Integer>();

        // Add values to the set
        numbers.add(4);
        numbers.add(7);
        numbers.add(8);

        // Show which numbers between 1 and 10 are in the set
        for (int i = 1; i <= 10; i++) {
            if (numbers.contains(i)) {
                System.out.println(i + " was found in the set.");
            } else {
                System.out.println(i + " was not found in the set.");
            }
        }
    }
}
```

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The var Keyword

From Java 10, you can use the **var** keyword to declare a **HashSet** variable without writing the type twice. The compiler figures out the type from the value you assign.



Example

```
// Without var
HashSet<String> cars = new HashSet<String>();

// With var
var cars = new HashSet<String>();
```

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The Set Interface

Note: Sometimes you will see both `Set` and `HashSet` in Java code, like this:

```
import java.util.Set;
import java.util.HashSet;

Set<String> cars = new HashSet<>();
```

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This means the variable (cars) is declared as a `Set` (the interface), but it stores a `HashSet` object (the actual set). Since `HashSet` implements the `Set` interface, this is possible.

It works the same way, but some developers prefer this style because it gives them more flexibility to change the type later.

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Exercise [?]

What is a correct syntax to create a HashSet object that will store strings?

- ☐ `Hash<String> myObj = new HashSet<String>();`
- ☐ `HashMap<String> myObj = new HashMap<String>();`
- ☐ `HashSet<String> myObj = new HashSet<String>();`
- ☐ `HashMap<String> myObj = new HashSet<String>();`

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